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Source:

CSE400 – Fundamentals of Probability in Computing

Lecture 7: Expectation, CDFs, PDFs and Problem Solving

Instructor: Dhaval Patel, PhD
Date: January 27, 2025

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1. Cumulative Distribution Function (CDF)

1.1 Definition of CDF

Let X be a random variable.

$$F_X(x) = \Pr(X \leqslant x), \quad -\infty < x < \infty$$

- $F_X(x)$ denotes the **Cumulative Distribution Function (CDF)** of random variable X.
- Most of the information about the random experiment described by the random variable X is determined by the behavior of $F_X(x)$.

1.2 Properties of CDF

For a valid CDF $F_X(x)$, the following properties must hold:

1. **Boundedness**

$$0 \leqslant F_X(x) \leqslant 1$$

2. **Limits at infinity**

$$F_X(-\infty) = 0$$

$$F_X(\infty) = 1$$

3. **Monotonicity**
   For $x_1 < x_2$,

$$F_X(x_1) \leqslant F_X(x_2)$$

(Non-decreasing / monotonic increasing)

4. **Probability over an interval**
   For $x_1 < x_2$,

$$\Pr(x_1 < X \leqslant x_2) = F_X(x_2) - F_X(x_1)$$

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2. CDF – Example #1

Problem:

Find the valid CDF

(a)

$$F_X(x) = \frac{1}{2} + \frac{1}{\pi}\tan^{-1}(x)$$

- Satisfies:
  - $0 \leqslant F_X(x) \leqslant 1$
  - $F_X(-\infty) = 0$
  - $F_X(\infty) = 1$
  - Monotonic increasing

✔️ **Valid CDF**

(b)

$$F_X(x) = [1 - e^{-x}]\, u(x)$$

- $u(x)$: Unit Step Function
- Defined as:

$$u(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geqslant 0 \end{cases}$$

✔️ **Valid CDF**

(c)

$$F_X(x) = e^{-x^2}$$

✘ **Invalid CDF**
(Does not satisfy required CDF properties)

(d)

$$F_X(x) = x^2 u(x)$$

✘ **Invalid CDF**

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3. CDF – Example #2

Given:

$$F_X(x) = (1 - e^{-x})u(x)$$

Find:

(a) $\Pr(X > 5)$

$$\Pr(X > 5) = 1 - \Pr(X \leqslant 5)$$

$$= 1 - F_X(5)$$

$$= e^{-5}$$

(b) $\Pr(X < 5)$

$$\Pr(X < 5) = F_X(5)$$

(c) $\Pr(3 < X < 7)$

$$\Pr(3 < X < 7) = F_X(7) - F_X(3)$$

(d) $\Pr(X > 5 \mid X < 7)$

Let:

- $A = \{X > 5\}$
- $B = \{X < 7\}$

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}$$

$$= \frac{\Pr(5 < X < 7)}{\Pr(X < 7)}$$

$$= \frac{F_X(7) - F_X(5)}{F_X(7)}$$

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4. Probability Density Function (PDF)

4.1 Definition of PDF

For a **continuous random variable**, the PDF of X evaluated at point x is:

$$f_X(x) = \lim_{\epsilon \to 0} \frac{\Pr(x \leqslant X < x + \epsilon)}{\epsilon}$$

4.2 PDF–CDF Relationship (Derivation)

Recall for a continuous range:

$$\Pr(x \leqslant X < x + \epsilon) = F_X(x + \epsilon) - F_X(x)$$

Substitute into PDF definition:

$$f_X(x) = \lim_{\epsilon \to 0} \frac{F_X(x + \epsilon) - F_X(x)}{\epsilon}$$

Taking the limit:

$$\boxed{f_X(x) = \frac{dF_X(x)}{dx}}$$

4.3 Important Result

- The **PDF of a random variable is the derivative of its CDF**
- Conversely, the **CDF of a random variable is the integral of its PDF**

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\, dt$$

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5. Relationship Between PDF and CDF

- Area under the PDF curve from $-\infty$ to x gives $F_X(x)$
- Graphical relationship shown in the slide:
  - Shaded area under $f_X(x)$ up to x_0 equals $F_X(x_0)$

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6. Expectation of Random Variables

(Only headings present in slides – content continues in later slides)

Topics listed:

- Definition and Example
- Expectation of a Function of RV
- Linear Operation with Expectation
- n^{th} Moments and Central Moments of RVs:

- Variance
- Skewness
- Kurtosis

(Details appear in subsequent slides beyond the provided content)

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